

Data Science Workflow – Exploration with Julius AI

Step 0 (Optional): Know Your Data

Purpose: If the dataset is unfamiliar, take a moment to see what it contains before you decide what to do.

Sometimes you already know what kind of data you're using.

But if it's new, this short step helps you avoid wrong assumptions later.

Take a quick look:

- What kind of things does it describe? (customers, products, campaigns, sales)
- How big is it — a few hundred or thousands of entries?
- Do the column names sound familiar or confusing?
- Does anything immediately look off or strange?

Example prompts to ask in Julius AI:

"Give me a short summary of this dataset — what does it seem to describe? Maybe mention a few interesting columns if they stand out." "List all column names with short explanations."
"What time period or topic does this dataset appear to cover?"

At this point, you are only **observing**, not deciding.

You want a general sense of what kind of information you have, not what to do with it yet.

Step 1: Define the Goal

Purpose: Turn curiosity into a question you can actually answer.

Every good analysis starts with curiosity — but curiosity alone isn't enough.

You need to decide what you want to learn or improve.

Ask yourself:

- What do I want to understand or decide?
- Who could use this information?
- What would success look like if I found an answer?

Examples:

- "Which marketing efforts really pay off?"
- "When are customers most active?"
- "What makes people come back?"
- "Is my pricing consistent across products?"

Example prompts in Julius AI:

"Help me turn this question into a measurable goal: which campaigns perform best?"
"What metrics could describe customer loyalty in this dataset?" "Based on this data, which metric would make the most sense to focus on first — CLV, ROI, Retention, Churn, or Conversion? Please explain briefly why it fits"

By defining your goal clearly, every later step becomes easier and more focused.

Step 2: Understand the Data

Purpose: Explore what matters, what's redundant, and what might need attention.

Now that you have a goal, it's time to understand the dataset in more depth.
This is about interpreting what you see, not just reading it.

Focus on:

- Which columns seem **relevant** for your goal, and which don't
- Which ones are **key identifiers** (like IDs or dates)
- Which describe **behavior or results** (like amount spent, duration, score)
- Where the data looks **incomplete, inconsistent, or irrelevant**

Example prompts in Julius AI:

"Show a quick summary of each column — what it represents and how complete it is"
"Which columns seem unrelated to my goal of understanding campaign success?"
"Are there values or ranges that look unrealistic?"
"Which features seem most informative for analysis?" "For the metric we picked, which tables and columns seem most relevant? Please describe how they connect and if anything important might be missing or unclear"

This is the step where you start making first judgments:

- Are some columns safe to ignore because they don't add value?
- Are there duplicated or overlapping fields?
- Could some outliers be real signals worth keeping?

You're still not cleaning yet — but you are **deciding what deserves your attention**.

Step 3: Clean & Prepare

Purpose: Make the dataset consistent, usable, and trustworthy.

After understanding what's in the data, the next step is to fix or adapt it.
Here you apply the decisions from Step 2: what stays, what goes, what needs correction.

Typical actions:

- Remove or flag data that's clearly wrong or irrelevant
- Handle missing values (fill, drop, or estimate based on context)
- Standardize formats (dates, currencies, categories)

- Review outliers — keep if meaningful, remove if they distort results
- Make sure keys like customer IDs or product SKUs are unique

Example prompts in Julius AI:

"Find and list inconsistent date or number formats."
"Highlight outliers that could influence averages."
"Suggest ways to handle missing values in this dataset."
"Which columns should be cleaned or standardized before analysis?"

This step is both technical and interpretive —

AI can detect issues, but **you decide what's correct**.

Good cleaning is what separates reliable insights from false conclusions.

Step 4: Explore and Analyze

Purpose: Discover relationships and patterns that connect back to your goal.

Now the data is ready — clean, consistent, and familiar.

This is where curiosity becomes analysis.

Focus on:

- Finding **patterns**: do certain behaviors or features repeat?
- Checking **relationships** between variables
- Looking for **trends over time**
- Comparing **groups** (e.g. customer segments, campaigns, products)

Typical exploration questions:

- Do loyal customers spend more?
- Which campaigns bring higher engagement?
- Are there seasonal effects on sales?
- Does income or region influence customer type?

Example prompts in Julius AI:

"Show the correlation between customer lifetime value and campaign type."
"Visualize spending by age group and region."
"Find any interesting trends over time."
"Compare engagement between returning and new customers."

At this stage, the goal is *to notice connections*, not to build complex models.

Exploration helps you see what deserves deeper attention later.

Step 5: Visualize

Purpose: Turn numbers into insights that people can understand.

Data doesn't speak for itself — it needs to be seen.

Visualization is where findings become stories.

Focus on:

- Showing **relationships** (scatter plots, heatmaps, line charts)
- Highlighting **comparisons** (bars, grouped metrics)
- Illustrating **change over time** (time series, rolling averages)
- Using **simple, clear visuals** that anyone can interpret

Example prompts in Julius AI:

"Create a simple bar chart showing ROI by campaign type."

"Plot customer retention over time."

"Visualize the top 5 products by total revenue."

"Summarize key insights from this visualization."

Good visualization answers the original question in a form that can be shared and discussed. It also helps you notice what's missing — which often leads to new questions.

Step 6: Interpret and Decide

Purpose: Bring meaning to results — what do they tell you, and what comes next?

After visualization comes interpretation: understanding what your findings actually imply.

Ask yourself:

- Do the results make sense in context?
- Are there alternative explanations?
- What new questions or decisions emerge?
- What would you test or check next?

Example prompts in Julius AI:

"Summarize what these results might imply for campaign strategy."

"Which insights seem most relevant for customer retention?"

"What additional data might help confirm these findings?"

"Suggest possible next steps based on these trends."

AI can help you summarize, compare, and highlight. But deciding *what matters* and *what to do* with it — that's still human work.

Step 7: Iterate and Refine

Purpose: Use what you've learned to improve both your questions and your data.

Data work is rarely linear. Each analysis reveals something that suggests the next step.

Focus on:

- Reviewing your assumptions — were they right?
- Refining the goal — do you see a clearer question now?
- Updating your dataset — do you need more data or better quality?
- Documenting insights — so others can follow your reasoning

Example prompts in Julius AI:

"Based on these insights, suggest follow-up questions."
"Which variables deserve deeper analysis?"
"Identify what additional data could improve this model."
"Summarize lessons learned from this exploration."

Iteration turns analysis into learning.

Every loop makes your understanding — and your results — stronger.

Closing Thoughts

Purpose: Connect the workflow back to real-world decisions.

The workflow isn't about tools or code.

It's a way of thinking:

1. Start with curiosity.
2. Understand what your data really says.
3. Prepare it carefully.
4. Explore, visualize, and interpret with intent.
5. Use AI as a guide — not as an answer.

When done right, the process builds not just insights,
but confidence in how you use data to make decisions.